Algorithms for Skin Cancer Detection Using Deep Learning

How They Work, Published Results, and the Recommended Pipeline

Project: Skin Cancer Detection Using Deep Learning

Companion document to the IEEE paper

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Abstract

This document explains the algorithms used for skin cancer detection from images, from classical machine learning to convolutional neural networks (CNNs), vision transformers, hybrid models, ensembles, and dermatology foundation models, together with the U-Net family for lesion segmentation. For each family it describes the core idea, model size, strengths, and weaknesses. It then compares published results. The results are grouped by evaluation protocol, because numbers obtained under different datasets and data splits are not directly comparable. The document ends with the recommended pipeline for this project and the evidence behind each choice. All numbers are quoted exactly as reported in the cited papers.

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1 The Task

Skin cancer detection from images involves two computer-vision tasks:

- **Classification:** given an image of a lesion, predict its diagnosis. It can be *binary* (malignant vs. benign, or melanoma vs. rest) or *multi-class*: 7 classes in HAM10000 [1] and 8 in ISIC 2019 (melanoma, BCC, SCC, actinic keratosis, nevus, benign keratosis, dermatofibroma, vascular lesion).
- **Segmentation:** outline the lesion pixel by pixel, which helps the classifier focus on the lesion and allows border and size measurements.

Performance is measured with accuracy, balanced accuracy (BA, the average recall over classes), sensitivity and specificity for malignancy, ROC-AUC, F1-score, and (for segmentation) the Dice coefficient and Intersection-over-Union (IoU). Because the classes are highly imbalanced, BA is more informative than plain accuracy.

2 Algorithm Families

2.1 Classical Machine Learning

Idea. Features (colour histograms, texture, border shape, or raw resized pixels) are extracted and fed to a classical classifier such as a support vector machine (SVM), random forest, or k-nearest neighbours. **Strengths:** fast and simple, and handcrafted features are interpretable (e.g., the ABCD rule). **Weaknesses:** features must be designed by hand and do not capture complex visual patterns. **Evidence:** on the same HAM10000 subset, an SVM reached 53% and a random forest 87% accuracy, while a small CNN reached 94% [2]. **Use:** baseline only.

2.2 Convolutional Neural Networks (CNNs)

A CNN learns its own features. Stacked convolution layers slide small filters over the image, detecting edges first, then textures, then lesion structures such as pigment networks and globules. Pooling layers reduce resolution, and a final fully connected layer outputs class probabilities. Almost all skin cancer CNNs use *transfer learning*: they start from weights pre-trained on ImageNet and are fine-tuned on dermatology images, which is essential because dermatology datasets are small.

VGG16/19 [3]. A plain stack of 3×3 convolutions. It is easy to understand, but with 138M parameters it is heavy for its accuracy. It is now mainly a historical baseline.

ResNet [4]. Adds *residual (skip) connections*, $y = F(x) + x$, which let very deep networks (50–152 layers) train without vanishing gradients. ResNet-50 (25.6M parameters) is the most common baseline. ResNeXt-101 was the best of five CNNs on HAM10000 (93.2%) [5].

Inception / Xception [6, 7]. Use parallel filters of different sizes (Inception) or depthwise-separable convolutions (Xception) to capture multi-scale patterns efficiently. InceptionV3 was used in the landmark dermatologist-level study [8]. InceptionResNetV2 with soft attention reached 93.4% on HAM10000 [9].

DenseNet [10]. Connects every layer to all later layers, so features are reused. DenseNet-121 has only 8.0M parameters. DenseNet201 reached a weighted AUC of 0.974 on HAM10000 [9].

MobileNetV2 [11]. Uses inverted residual blocks with depthwise convolutions (3.5M parameters), designed for phones. A MobileNet reached 83.1% on HAM10000 [12]. It is a good choice for on-device apps.

EfficientNet / EfficientNetV2 [13, 14]. Scale depth, width, and input resolution together with one *compound* coefficient (B0–B7). They give the best accuracy per unit of compute. EfficientNet ensembles won both the ISIC 2019 [15] and SIIM-ISIC 2020 [16] challenges. B4 gave the best single result among B0–B7 on HAM10000 (87.9%) [17]. EfficientNetV2 trains faster by using fused convolution blocks in early layers.

ConvNeXt [18]. A "modernized" ResNet that borrows design ideas from transformers (large 7×7 kernels, LayerNorm, GELU, fewer activations). ConvNeXt-T (28M) matches Swin accuracy with CNN efficiency. An ensemble of four ConvNeXt models achieved 84.2% mean sensitivity on ISIC 2019 [19].

2.3 Vision Transformers

ViT [20]. Splits the image into 16×16 patches, embeds each patch as a token, and applies *self-attention*, so every patch can attend to every other patch:

$$\text{Attention}(Q, K, V) = \text{softmax}\Big(\frac{QK^\top}{\sqrt{d_k}}\Big)V. \tag{1}$$

This captures global context, such as asymmetry across the whole lesion, but ViTs need large training sets because they lack the built-in locality of CNNs. The SkinTrans ViT reported 94.3% on HAM10000 [21].

Swin Transformer [22]. Computes attention inside local windows that *shift* between layers, producing a CNN-like hierarchy of feature maps at lower cost. Swin reached a balanced accuracy of 82.3% on the eight-class ISIC 2019 set [23].

2.4 Hybrid CNN–Transformer Models

Hybrids combine a CNN (local texture, colour, borders) with a transformer (global structure). With the same data split, a ResNet stem followed by a ViT reached 89.5% on ISIC 2018, versus 83.7% for the CNN alone [24]. Other examples fuse Swin with a DenseNet attention branch (94.2% on HAM10000) [25], or ConvNeXt with ViT through attention (MedFusionNet) [26]. **Weakness:** most hybrids are evaluated only on one benchmark, without external testing.

2.5 Ensembles

An ensemble averages the predictions of several models. If the models make *different* errors, the average is more accurate and better calibrated than any member. Every recent ISIC challenge was won by an ensemble [15, 16]. On ISIC 2019, a single EfficientNet-B0 reached 65.8% BA, and the full multi-resolution ensemble with metadata 74.2% (cross-validation) [15]. **Weakness:** inference cost grows with the number of models.

2.6 Foundation Models

Foundation models are pre-trained on millions of images, often without labels (self-supervised learning), and then adapted to many tasks.

- **PanDerm** [27] was pre-trained on more than 2 million dermatology images from 11 institutions. It achieved state-of-the-art results on 28 benchmarks and improved clinicians' skin cancer accuracy by 11%.
- **DINOv2** is a general self-supervised vision model. Its frozen features are a strong, cheap baseline.
- **MONET** [28] links images to medical concepts (e.g., "pigment network") for auditing models.
- **SkinGPT-4** [29] adds a chat interface on top of an image encoder.

Weakness: they are large (ViT-L, about 300M parameters) and compute-heavy, and some weights have licence restrictions.

2.7 Segmentation Networks

U-Net [30]. An encoder–decoder with skip connections that copy high-resolution features to the decoder. It is the standard for medical segmentation.

Attention U-Net [31]. Adds attention gates to the skip connections that suppress irrelevant background such as healthy skin, hair, or rulers.

TransUNet [32]. Uses a transformer in the encoder for global context.

UCTransNet [33]. Replaces plain skip connections with channel-wise transformer modules.

2.8 Supporting Techniques

- **Focal loss** [34]: $\mathscr{L} = -\alpha_t(1-p_t)^\gamma \log p_t$ down-weights easy, frequent examples so that the model concentrates on rare cancers. It added 9.5 accuracy points to a ResNet-50 under identical conditions [24] (Fig. 3).
- **Data augmentation and synthetic images**: flips, rotations, colour jitter, and CutMix are standard. GANs and diffusion models can synthesize rare lesions; synthetic melanomas raised AUC from 0.925 to 0.945 in one study [35].
- **Metadata fusion**: adding age, sex, and body site to the image features gave consistent gains in the ISIC 2019 winning solution [15].
- **Grad-CAM** [36]: produces a heatmap of the image regions that drove a prediction. It is the most common explanation tool, although its effect on clinicians is rarely measured [37].

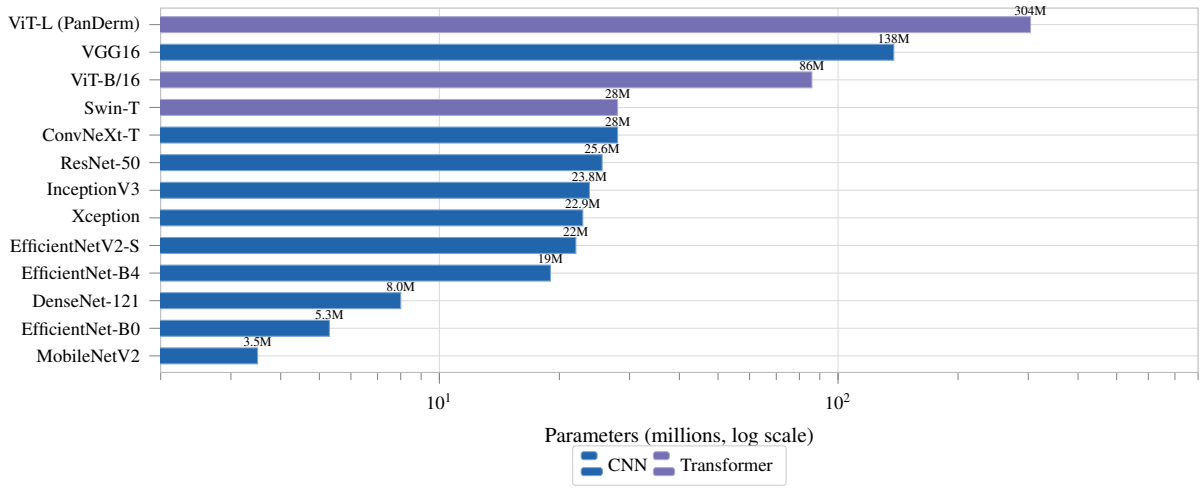


Figure 1: Model size of standard ImageNet backbones. Models below about 10M parameters run comfortably on phones; transformers and VGG need a server GPU.

3 Comparison of Published Results

Different papers use different datasets, class sets, splits, and metrics, so the results are grouped into **four comparable panels**. Within a panel the conditions are similar. Numbers from different panels should not be compared directly.

Table 1: Published results, exactly as reported in each source.

Panel	Family	Model	Dataset	Result	Protocol
A	Classical ML	SVM [2]	HAM10000 subset	Acc. 53%	*k*-fold, augmented subset
A	Classical ML	Random forest [2]	HAM10000 subset	Acc. 87%	same
A	CNN	ResNet-50, CE [24]	ISIC 2018	Acc. 74.2%	random 70/10/20
A	CNN (light)	MobileNet [12]	HAM10000	Acc. 83.1%	random split
A	CNN	ResNet-50, focal [24]	ISIC 2018	Acc. 83.7%	same as CE row
A	CNN	EfficientNet-B4 [17]	HAM10000	Acc. 87.9%, F1 0.87	best of B0–B7
A	Hybrid	ResNet + ViT, focal [24]	ISIC 2018	Acc. 89.5%	same as CE row
A	CNN	ResNeXt-101 [5]	HAM10000	Acc. 93.2%	best of five CNNs
A	CNN + attention	IRv2 + soft attention [9]	HAM10000	Acc. 93.4%, AUC 0.984	resampled split
A	Hybrid	Swin + DenseNet [25]	HAM10000	Acc. 94.2%	single dataset
A	Transformer	SkinTrans (ViT) [21]	HAM10000	Acc. 94.3%	split not stated
B	CNN	EfficientNet-B0 [15]	ISIC 2019	BA 65.8%	5-fold CV
B	CNN	EfficientNet-B0 + metadata [15]	ISIC 2019	BA 67.4%	5-fold CV
B	Ensemble	Multi-res. EfficientNets + meta [15]	ISIC 2019	BA 74.2%	5-fold CV
B	Ensemble	same, official test [15]	ISIC 2019 test	BA 63.6%	hidden test
B	Transformer	Swin [23]	ISIC 2019	BA 82.3%	hold-out
B	Ensemble	ConvNeXt T/S/B/L [19]	ISIC 2019	Sens. 84.2%	hold-out
C	ViT (ImageNet)	ViT-L linear probe [27]	HAM10000 clean	BA 65.3%, AUC 0.970	patient-disjoint
C	Self-supervised	DINOv2 ViT-L probe [27]	HAM10000 clean	BA 70.1%, AUC 0.964	patient-disjoint
C	Foundation	PanDerm ViT-L probe [27]	HAM10000 clean	BA 80.7%, AUC 0.988	patient-disjoint
D	CNN	U-Net [38]	ISIC 2018 T1	Dice 0.867, IoU 0.849	uniform re-run
D	CNN + attention	Attention U-Net [38]	ISIC 2018 T1	Dice 0.882, IoU 0.865	same
D	Transformer	TransUNet [38]	ISIC 2018 T1	Dice 0.850, IoU 0.837	same
D	Hybrid	UCTransNet [38]	ISIC 2018 T1	Dice 0.890, IoU 0.873	same
–	Ensemble	EfficientNet ens. [16]	ISIC 2020	AUC 0.949	official leaderboard
–	19 CNNs	best after de-dup. [39]	ISIC 2020	AUC 0.80	duplicates removed

3.1 Panel A: Seven-Class HAM10000 / ISIC 2018

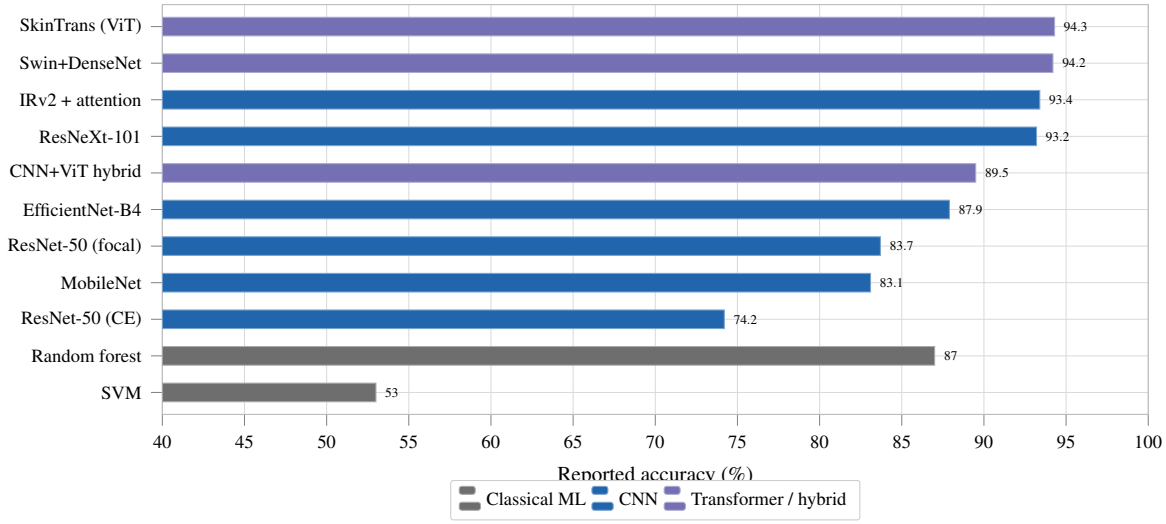


Figure 2: Panel A: seven-class accuracy (random splits, so values are optimistic [39]).

Deep learning clearly beats classical ML. Attention, hybrid, and transformer models lead with 89–94%. All of these papers split images at random, so the true accuracy on new patients is lower.

3.2 Controlled Evidence: Loss Function and Architecture

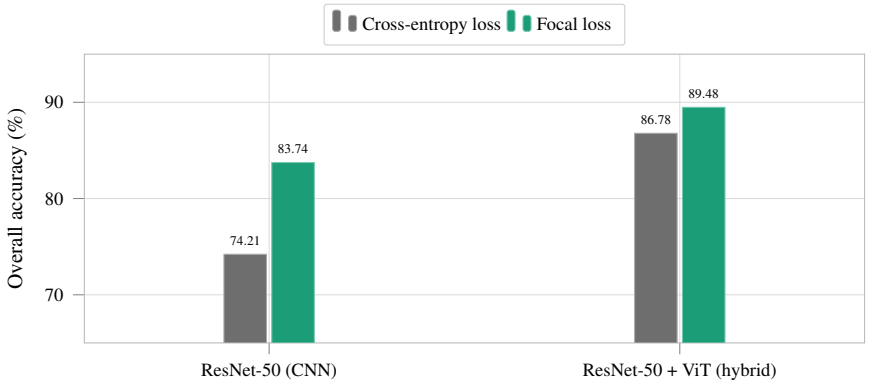


Figure 3: Same data and split, different loss and architecture (ISIC 2018) [24]. Focal loss and the hybrid transformer each add several points.

3.3 Panel B: Eight-Class ISIC 2019

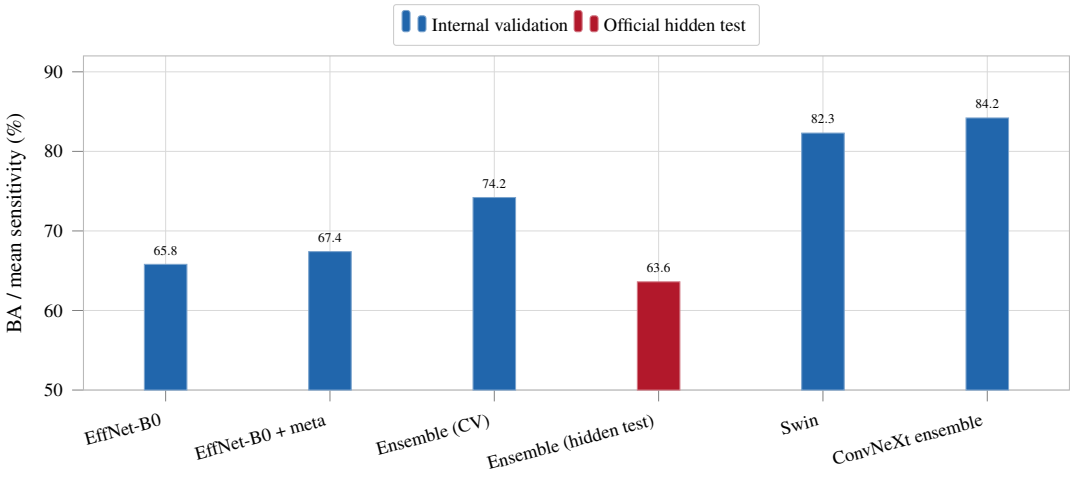


Figure 4: Panel B: ISIC 2019. The winning ensemble lost 10.6 points on the hidden test set [15, 19, 23].

Ensembles beat single models, and modern backbones (Swin, ConvNeXt) are the strongest. However, internal validation overestimates test performance by about 10 points, more than the gap between architectures.

3.4 Panel C: Patient-Disjoint Split (Fairest Comparison)

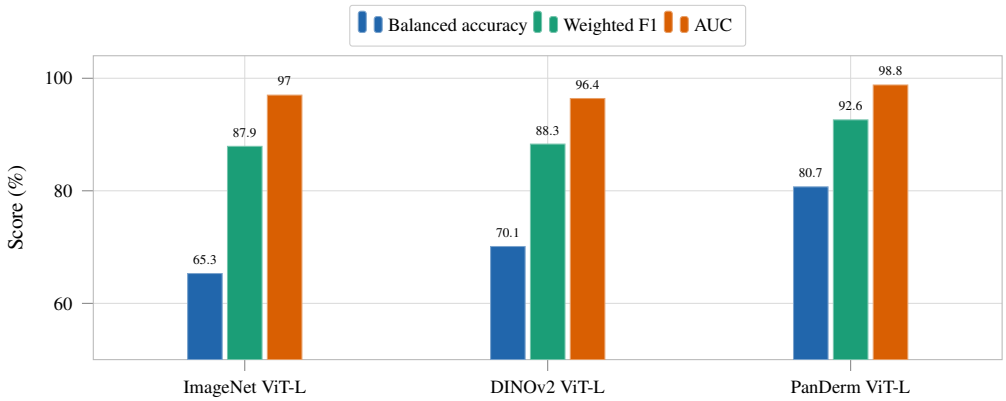


Figure 5: Panel C: identical frozen-encoder protocol on a patient-disjoint split [27]. Pre-training on dermatology data gives the biggest gain.

3.5 Panel D: Lesion Segmentation

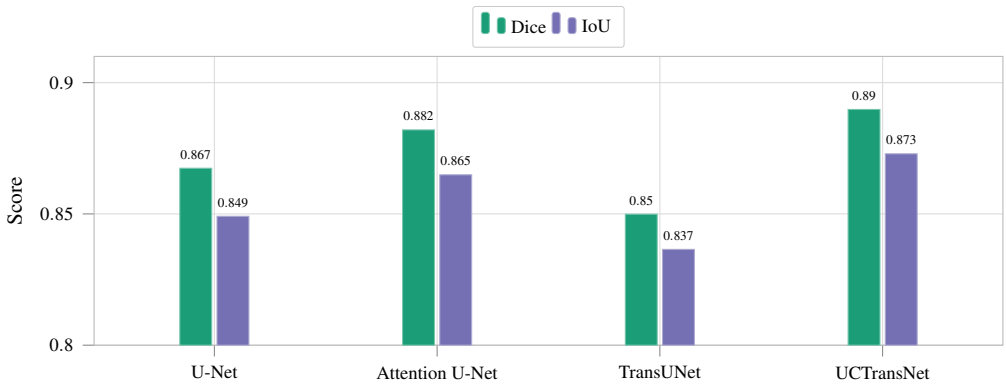


Figure 6: Panel D: ISIC 2018 Task 1, one uniform re-implementation [38].

3.6 From Benchmark to Clinic

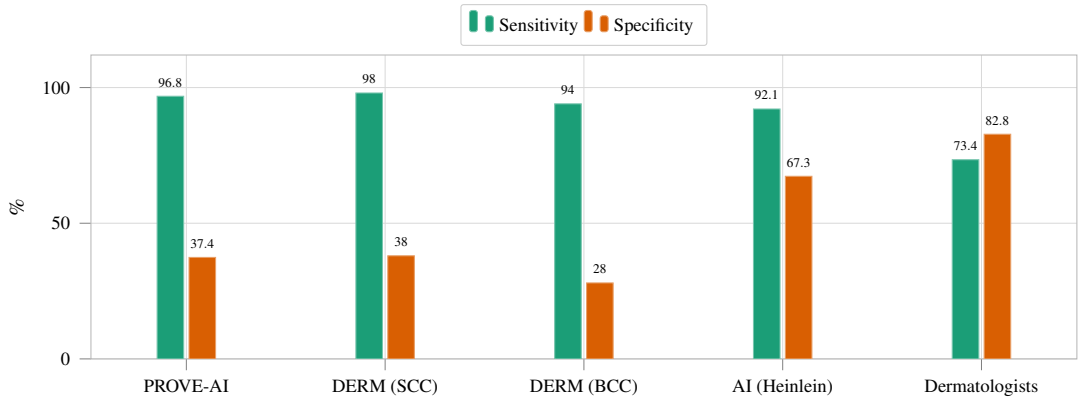


Figure 7: Prospective clinical studies [40–42]: AI achieves high sensitivity, but specificity is low, which means many unnecessary referrals.

4 Strengths and Weaknesses at a Glance

Table 2: Qualitative comparison of algorithm families.

Family	Examples	Params	Strengths	Weaknesses
Classical ML	SVM, random forest	<1M	Fast; interpretable features	Much lower accuracy
Plain CNN	VGG16, ResNet-50	25.6–138M	Mature; easy transfer learning	VGG heavy; little global context
Efficient CNN	DenseNet, MobileNetV2	3.5–8M	Small; phone-friendly	Weaker on rare classes
Scaled CNN	EfficientNet(V2)	5–22M	Best accuracy per FLOP; challenge winner	Sensitive to input size
Modern CNN	ConvNeXt	28M	Transformer-style design, CNN speed	Needs strong augmentation
Transformer	ViT, Swin	28–86M	Global context; strong on ISIC 2019	Data-hungry; memory at high resolution
Hybrid	ResNet+ViT, Swin+DenseNet	varies	Local + global features	Complex; rarely externally tested
Ensemble	Multi-backbone voting	sum	Most robust; best calibrated	Slower inference
Foundation	PanDerm, DINOv2	∼300M	Best transfer; few labels needed	Heavy compute; licences

5 Recommended Pipeline for This Project

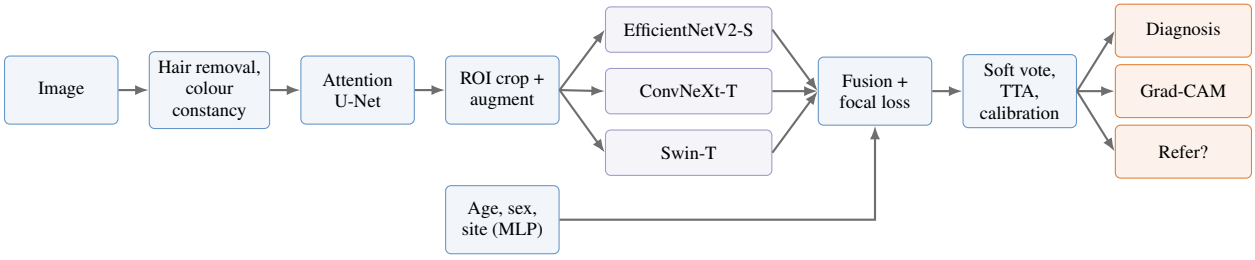


Figure 8: Recommended pipeline: segmentation-guided hybrid ensemble with metadata fusion.

Table 3: Recommended component for each stage and the evidence for it.

Stage	Choice	Evidence / reason
Segmentation	Attention U-Net (EfficientNet encoder)	Best U-Net variant under a uniform test (Dice 0.882 vs. 0.867) [38]
Backbone 1	EfficientNetV2-S	EfficientNets won ISIC 2019 and 2020 [15, 16]
Backbone 2	ConvNeXt-T	Best reported ISIC 2019 sensitivity (84.2%) [19]
Backbone 3	Swin-T	Transformer diversity; BA 82.3% on ISIC 2019 [23]
Imbalance	Focal loss ($\gamma = 2$) + balanced sampling	+9.5 points under identical conditions [24]
Metadata	MLP on age, sex, site	Consistent gains in the ISIC 2019 winner [15]
Combination	Weighted soft vote + TTA + temperature scaling	Ensembles win challenges; calibration makes probabilities usable
Decision	Threshold for $\geq 95\%$ malignant sensitivity	Prospective studies show specificity collapses otherwise [40]
Explanation	Grad-CAM	Standard, model-agnostic heatmaps [36]
Strong baseline	Linear probe on DINOv2 features	Cheap; 70.1% BA patient-disjoint [27]

5.1 Implementation Notes

- **Framework:** PyTorch with the `timm` library, which provides pre-trained EfficientNetV2, ConvNeXt, and Swin models (e.g., `tf_efficientnetv2_s`, `convnext_tiny`, `swin_tiny_patch4_window7_224`).
- **Segmentation:** the `segmentation_models_pytorch` library provides U-Net with an EfficientNet encoder, to which attention gates can be added.
- **Explanations:** the `pytorch-grad-cam` package.
- **Hardware:** one 16–24 GB GPU with mixed precision is enough. Train 30 epochs with AdamW and a cosine schedule, using 5-fold lesion-grouped cross-validation.
- **Baselines to report:** ResNet-50, DenseNet-121, MobileNetV2, and each single backbone under the same pipeline, so that the ensemble's gain is measured, not assumed.

Summary: deep learning clearly beats classical ML; ensembles of complementary CNN and transformer backbones, trained with focal loss and metadata, are the most effective practical choice. Evaluation choices (splits, external data, metric) change results as much as the architecture does. The recommended pipeline is Attention U-Net followed by an EfficientNetV2-S, ConvNeXt-T, and Swin-T ensemble with metadata, focal loss, calibration, and Grad-CAM.

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